

WASP-2b

R-2246

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_120914 · created 2026-08-15T23:30:18+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456184.6267 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.047
Transit depth (Rp/Rs)^2	2.56 +/- 1.49 [%]
Orbital Inclination (inc)	86.23 +/- 2.87
Airmass coefficient 1 (a1)	1.78 +/- 0.15
Airmass coefficient 2 (a2)	-0.52 +/- 0.5
Scatter in the residuals of the lightcurve fit is	7.83 %
Best Comparison Star	#2 - [190, 320]
Optimal Aperture	4.52
Optimal Annulus	8.45
Transit Duration (day)	0.043 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.047 vs 0.1326 (deviation 0.55σ)
Duration fit vs published	0.043 d vs 0.0737 d (deviation 2.08σ)
Mid-time O–C	-4.2 min
Depth SNR	3.2
Residual scatter	7.83 %

Light curve

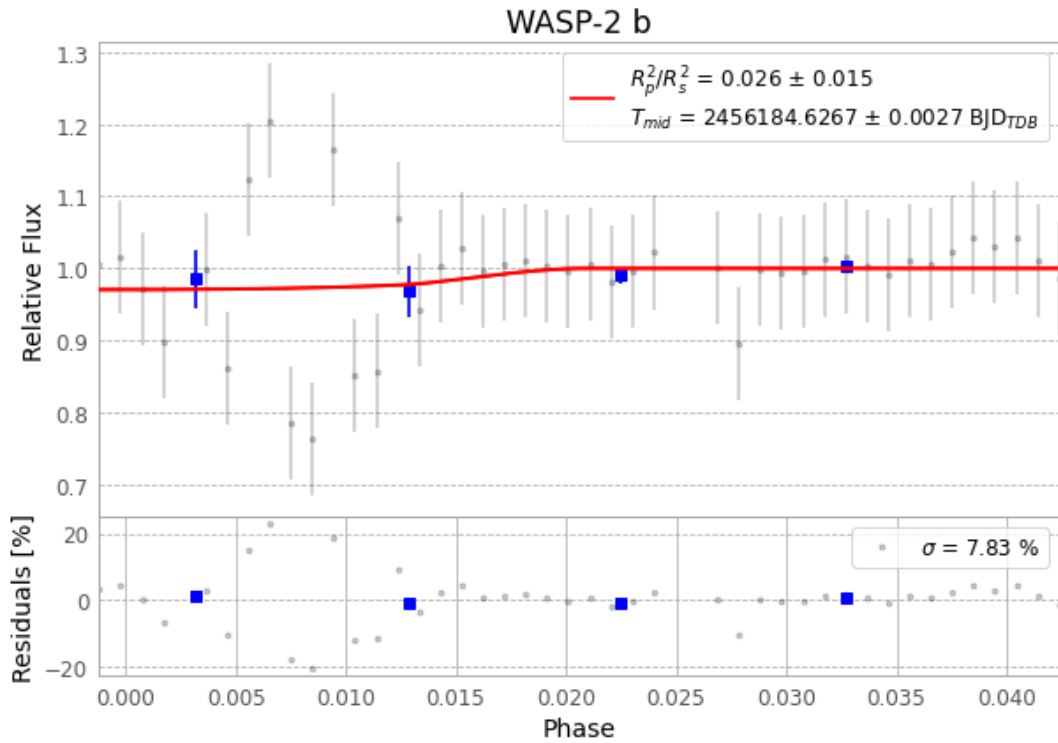


Figure 1 — FinalLightCurve_WASP-2 b_2012-09-13.png (SHA-256 4bd73689abb9d361...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [384, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 741d02aea1fde935...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

46 FITS frames (30 MB), WASP-2120914025112.FITS → WASP-2120914050612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-120914.log (3744bb382eeb...), FinalLightCurve_WASP-2 b_2012-09-13.png (4bd73689abb9...), FinalLightCurve_WASP-2 b_2012-09-13.csv (a1ae06938364...)

Compute resources

Wall time 133.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:30Z.